

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Table of Contents

1. Project Overview	1
2. Background & Business Objective	1
3. Product Purpose	2
4. System Functionalities	2
5. User Stories & Use Cases	3
6. Features Included (Scope Definition)	3
7. Features Not Included (Scope Control)	3
8. Assumptions & Constraints	4
9. Risks & Questions Throughout Development	4

Project Name: Greenloom

Version: 1.0

1. Project Overview

- **Problem Statement:** This project aims to replace manual reporting to ensure 100% compliance with Malaysia's 2026 Carbon Tax regulations while reducing their carbon emissions. However, many organizations struggle to accurately calculate their carbon emissions because relevant data is scattered across different operational areas such as transportation fuel usage, electricity consumption, and waste management. Traditional carbon accounting methods often require manual data collection and complex calculations, which are time-consuming, error-prone, and difficult to scale.
- **Target Domain:** The proposed system targets the environmental management and corporate sustainability domain. It is specifically designed for organizations that need to monitor and calculate their carbon emissions for internal sustainability reporting.
- **Proposed Solution Summary:** The proposed solution is an AI-powered organizational application designed to automatically collect environmental data, process carbon emission calculations, and generate carbon tax reports.

2. Background & Business Objective

- **Background of the Problem:** As governments implement carbon taxes to combat climate change, organizations are pressured to internalize environmental costs and adopt sustainable practices. However, carbon accounting remains hindered by manual processes and fragmented data—such as fuel, electricity, and waste records—scattered across various departments. This lack of integration prevents companies from gaining a clear, scalable view of their environmental impact and regulatory compliance.

- **Importance of Solving This Issue:** Addressing the challenges of carbon emission tracking is essential for organizations that aim to remain compliant with environmental regulations and demonstrate corporate social responsibility. Governments are increasingly introducing stricter carbon reporting requirements, making it essential for companies to maintain accurate emission records. Failure to comply with these requirements may lead to financial penalties, legal consequences, and reputational damage.
- **Strategic Fit / Impact:** The proposed AI-based carbon tax calculation system aligns with global sustainability initiatives and environmental governance frameworks. Many organizations are currently integrating Environmental, Social, and Governance (ESG) principles into their corporate strategies. Accurate carbon accounting plays a central role in ESG reporting, making the proposed system highly relevant to modern business operations. From a strategic perspective, the system enables organizations to transition from reactive compliance to proactive environmental management. By continuously monitoring emission sources such as fuel usage, electricity consumption, and waste generation, organizations can identify inefficiencies and implement greener operational practices.

3. Product Purpose

3.1. Main Goal of the System:

The primary goal of the proposed system is to provide organizations with a centralized platform for collecting environmental data, calculating carbon emissions, and generating carbon tax reports. The system aims to simplify the complex process of carbon accounting by automating data collection and applying AI-assisted analysis to improve accuracy and efficiency.

3.2. Intended Users (Target Audience):

- Company executives
- Procurement managers
- Operations staff
- System developers
- Sustainability analysts

4. System Functionalities

4.1. Description: The system functions as an AI-powered carbon governance platform that automates the transition from manual data entry to regulatory compliance. Leveraging AI vision for receipt extraction and waste estimation, the engine centralizes fragmented data across multiple emission scopes to calculate real-time tax liabilities. It integrates a policy-reading intelligence to align with current Malaysian regulations and a decision-support module that recommends low-carbon suppliers and tracks offsets, transforming raw operational data into audit-ready financial reports.

4.2. Key Functionalities:

- **Multi-Modal Data Ingestion**

It automates the transition from manual entry to digital precision. Utilizing AI Vision and OCR, the platform automatically extracts critical data from petrol receipts and utility bills to track emissions, while a Vision-based Wastage Estimator identifies waste categories and calculates weights directly from photos to bridge data gaps in facilities. This input layer is further enhanced by Logistics Automation, which synchronizes supplier routes to auto-calculate transportation distances and diesel consumption for comprehensive reporting.

- **AI Supplier Recommendation**

Analyzes staff chat data and supplier logistics to suggest low-carbon alternatives, proactively reducing footprint.

- **Regulatory Compliance**

A specialized module that ingests government PDF updates to automatically adjust internal tax rate simulations and compliance rules. It is also able to calculate how many carbon credits are needed to neutralize your tax liability.

- **Comprehensive Carbon Tax Reports**

The system is able to automatically generate a comprehensive carbon tax report that provides detailed insights into emissions sources, total emissions, and estimated tax liabilities. These reports can be used for regulatory compliance, sustainability monitoring, and organisational decision-making.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

The system uses a combination of AI technologies including computer vision, OCR, and natural language processing. Computer vision models analyse waste images to estimate weight and type. OCR extracts data from petrol receipts, while NLP structures transportation records for emission calculations. These models enable automated environmental data processing.

4.3.2. Prompting Strategy:

The prompts guide the AI to extract structured environmental data from the inputs. For waste images, prompts instruct the model to estimate waste type and weight. For petrol receipt, prompts focus on identifying fuel quantity, date, and fuel type to support carbon emission calculations.

4.3.3. Context & Input Handling:

The system accepts image uploads, text entries, and automated electricity data. Images are preprocessed for clarity before AI analysis. OCR converts receipt images into text, and structured electricity data is directly fed into the system to ensure consistent inputs for emission calculations.

4.3.4. Fallback & Failure Behavior:

If AI cannot confidently process inputs, the system triggers fallback procedures. Users may manually input waste weight or correct extracted receipt data. Error logging allows administrators to monitor AI performance and improve the models over time while maintaining calculation accuracy.

5. User Stories & Use Cases

- **User Stories:** A sustainability manager at an organization wants to monitor the company's environmental impact and calculate carbon tax obligations accurately. With the current system they have, they need to key in all data manually into the computer and their database.
- **Use Cases (Main Interactions):**

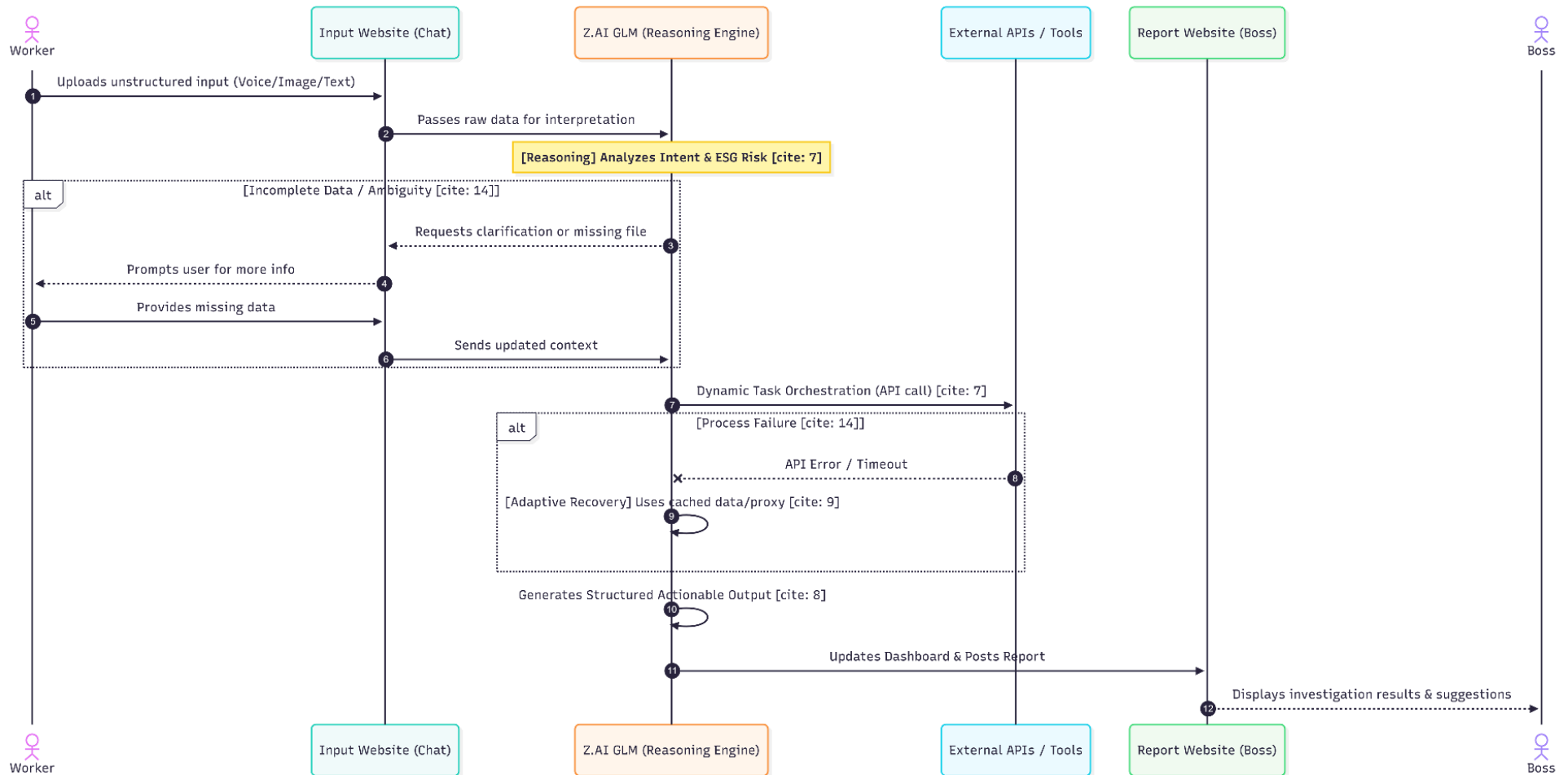
The main interaction within the system occurs when users submit environmental data and generate carbon tax reports. Employees begin by uploading transportation details and petrol receipts through the organization app. The system extracts relevant fuel consumption data and stores it in the database.

Electricity consumption data is automatically transmitted from the electric app to the

central system. Meanwhile, users can capture images of waste using the scanner function within the organization app. The AI model analyzes the image to estimate waste weight and categorize the waste type.

After sufficient data has been collected, authorized personnel such as sustainability managers can trigger the carbon emission calculation process. The system processes all input data, converts them into carbon emission values, and generates a comprehensive carbon tax report. This report can then be exported or shared with relevant departments for further action.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- (a) Centralised organisation application
Allows employees to submit transportation data and upload petrol receipts
Ensures that transportation-related emissions are accurately recorded and
Integrated into the carbon calculation process
- (b) Integration with electricity application
Automatically feeds electricity consumption data into the system
Eliminates the need for manual entry and ensures that energy usage data remains
accurate and up to date
- (c) Waste scanning function
Uses AI-powered image recognition to estimate the weight and category of waste
materials
Simplifies waste tracking and reduces reliance on manual measurement
- (d) Automated carbon emission calculations and report generation
Converts raw operational data into carbon emission estimates and calculates
the corresponding carbon tax liability
Provide organisations valuable insights into their environmental impact and
support regulatory compliance

7. Features Not Included (Scope Control)

- (a) Real-time integration with government carbon tax databases
- (b) Predictive carbon emission forecasting
- (c) Integration with external IoT waste sensors or smart electricity metres

8. Assumptions & Constraints

- (a) Organisations will encourage employees and logistics staff to consistently
upload accurate transportation data and petrol receipts.
- (b) Electricity application will provide accurate and consistent electricity
Consumption data.
- (c) The AI model used for waste image analysis can estimate waste weight
with reasonable accuracy.
- (d) Incomplete or inaccurate submissions may affect the quality of the final

carbon emission calculations.

9. Risks & Questions Throughout Development

- A Data Accuracy**

Since the platform relies on user-uploaded information and AI-generated estimates, there is a possibility that incorrect data may lead to inaccurate carbon tax calculations. This risk can be mitigated by implementing validation mechanisms and periodic data audits.
- B User Adoption**

Employees may be reluctant to consistently upload transportation records or waste images, especially if the process is perceived as time-consuming. To address this risk, the system should be designed with a simple and intuitive interface that minimizes the effort required for data submission.
- C Technical Risks**

If the waste recognition model fails to accurately estimate waste weight, the resulting emission calculations may become unreliable. Continuous model training and performance monitoring will be necessary to maintain system accuracy.